

Standard Operating Procedure (SOP) for Student Project Allocation, Execution & Monitoring

1. Purpose

This SOP defines a structured, data-driven, and outcome-oriented process for allocating, executing, monitoring, and evaluating **Machine Learning (ML)** in the Computer Engineering Department. The objective is to ensure algorithmic understanding, rigorous experimentation, regular progress tracking, and fair evaluation across semesters.

2. Scope

This SOP applies to:

- All undergraduate Computer Engineering students
 - All faculty members involved in project mentoring and evaluation
 - All project-based courses focusing on ML and DL from the 3rd semester onwards
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3. Project Models

3.1 Common Project Model (Phase 1: Weeks 1-10)

Objective:

- Build a strong foundation in the ML pipeline (Data → Model → Deployment)
- Ensure uniform learning outcomes regarding standard algorithms and metrics.
- Enable easier mentoring, monitoring, and benchmarking

Procedure:

1. The department shall define **common project problem statement** for all students.
2. **5 Problem Statement is given to student.**
3. Students must work individually as per course policy, but:
 - Dataset Source
 - Target Variables
 - Functional requirements shall remain common for all.
4. Mandatory Constraint: Implementation of at least **one algorithm without** the use of a library (scratch implementation).

Expected Outcomes:

- Clear understanding of Data Pre-processing, Feature Engineering, and EDA.
 - Hands-on experience with Model Training, Validation, and Hyperparameter Tuning.
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- Development of a Flask-based Web Interface for prediction.

Use any one dataset from following list:

- 1) Cardiovascular Disease Dataset (Healthcare) -
<https://www.kaggle.com/datasets/sulianova/cardiovascular-disease-dataset>
- 2) Bank Marketing Campaign (Marketing) -
<https://archive.ics.uci.edu/dataset/222/bank+marketing>
- 3) Vehicle Insurance fraud data –
<https://support.minitab.com/en-us/datasets/getting-started-guide-data-sets/insurance-fraud-data/>
- 4) Loan Default Prediction (Banking)
<https://www.kaggle.com/datasets/nikhil1e9/loan-default>
- 5) Stock market Price prediction
Use any free API to get data.

3.2 Deep Learning Project Model (Phase 2: Weeks 11-14)

Objective:

- Promote independent problem-solving in the Deep Learning domain.
- Encourage creativity, ownership, and exploration of unstructured data
- Prepare students for internships and industry projects

Procedure:

1. Each student shall select or propose an individual project title.
2. Project proposals must focus on unstructured data.
3. Recommended Technology Stack: PyTorch,
4. Minimum technical expectations include Custom Architecture Design (CNN/RNN)
5. Use Following dataset for ML task:

Use any one dataset from following list:

- 1) Face Mask Detector
[-https://www.kaggle.com/datasets/andrewmvd/face-mask-detection/data](https://www.kaggle.com/datasets/andrewmvd/face-mask-detection/data)

- 2) Traffic Sign Recognition
<https://idd.insaan.iiit.ac.in/dataset/details/>
- 3) Vehicle Log System (License Plate Recognition)–
<https://www.kaggle.com/datasets/saisirishan/indian-vehicle-dataset>

Expected Outcomes:

- Improved analytical and design skills regarding Neural Networks.
- Confidence in handling real-world problems and projects.

4. Project Planning & Weekly Execution

4.1 Week-wise Planning

At the **start of the semester**, a **week-wise project plan with project document** shall be shared with students.

Week No.	Title	Task List
1	Problem Definition and Dataset Exploration	Problem statement, dataset summary, and initial observations.
2	Data Cleaning and Pre-processing	Handle missing values Identify and handle outliers. Encode categorical variables Normalize/scale numerical features.EDA
3	Model Creation	Find out appropriate algorithm for training for your dataset. Use Library for project also Implement selected Algorithm without use of Library.
4	Model Evaluation	Test the model on the test dataset and compute metrics.Check for overfitting or underfitting.
5	Advanced Model Training	Experiment with advanced models cross-validation to ensure model stability. Compare models based on validation metrics. Hyperparameter Tuning
6	Visualization of metrics and graph	Display all types of graph associated with performance metrics

7	Flask Project Setup	Learn How to setup project on Flask Create simple flask application which use form .
8	Create Front end	Create front end of your project. Generally for user to input data
9	Create backend, Deployment	Setup backend to handle user request , Final Project Deployment to available free hosting site.
10	Project Evaluation	Project Evaluation
11	Find proper data for CNN project	Problem statement, dataset summary, and initial observations.
12	Setup Simple CNN architecture	Do proper setup using Pytorch
13	Final Deployment	Setup backend to handle user request , Final Project Deployment to available free hosting site.
14	Project Evaluation	Project Evaluation

5. Student Follow-up & Monitoring Mechanism

5.1 Regular Follow-ups

1. Faculty mentors shall conduct weekly project reviews. Each review shall verify:
2. Each review shall verify:
 - Code progress
 - Database changes
 - UI/screens completion
 - Alignment with weekly goals
3. Students must demonstrate:
 - Running code (not only screenshots)
 - Incremental commits or version updates

5.2 Progress Tracking

1. Students shall maintain:
 - Weekly progress records

- Task completion status
 - Delays or deviations must be Recorded, Communicated, and Corrected with clear action points.
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6. Roles & Responsibilities

6.1 Student Responsibilities

- Follow the defined project timeline strictly
- Attend all scheduled reviews
- Ensure originality and academic integrity
- Complete deliverables within deadlines

6.2 Faculty Mentor Responsibilities

- Guide students technically and conceptually
- Conduct regular reviews and provide feedback
- Ensure students adhere to SOP and standards
- Escalate issues if progress is unsatisfactory

6.3 Department Responsibilities

- Define project themes and expectations
 - Ensure uniform evaluation criteria
 - Review overall progress periodically
 - Update SOP as per academic or industry needs
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7. Evaluation Guidelines

7.1 Continuous Assessment Requirement:

- Evaluation must be performed iteratively throughout the project lifecycle. All follow-up sessions are to be logged in the Common Tracking Sheet.
- For common projects (ML), comparative benchmarking may be applied.
- For individual projects (DL), complexity and completeness shall be considered.
- Sheet template must align as below columns:
[Sr, Roll No, Enrolment No, Division, Batch, Student Name, Project Title (In case of common project this one will be optional), Evaluator Name(Taken By), Follow-up Date, Grade, Performance Remarks]

7.2 Grade distribution may include:

- Weekly progress & reviews: 1 Mark per objective.
- Code quality & architecture: Proper modularity and logic.
- Database/Dataset knowledge: Understanding of features and preprocessing.
- UI/UX consistency: Clean interface for prediction.
- Final demonstration & viva.

Evaluation Grading Criteria

Grade	Performance Level	Description
A	Very Good	Everything is perfect. The student clearly understands the project, and their work is neat, organized, and complete.
B	Good	Great effort. They understand the main parts of the project, but there are still a few small things that need to be fixed or improved.
C	Average	Just doing the work to get it done. They are finishing tasks but don't really understand how the code works or why they are doing it.
Z	Poor	Not doing the work. They are not following the project rules and haven't shown any interest in starting.

8. Compliance & Review

- Adherence to this SOP is **mandatory** for all students and faculty.
- Non-compliance may result in:
 - Reduced evaluation scores
 - Rework instructions
- This SOP shall be reviewed **every semester** by the department.

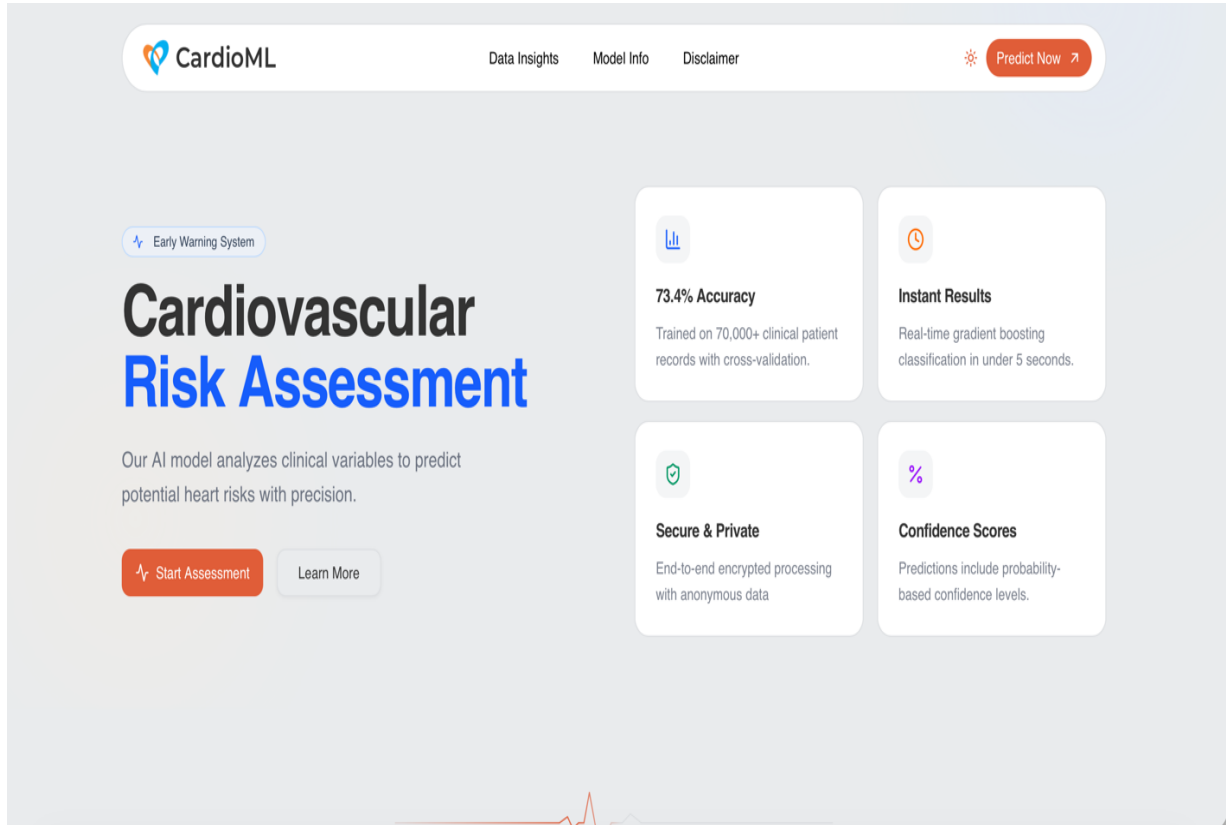
Effective From: Academic Year 2025–26

Approved By:

Prof. Rupesh G Vaishnav
(Head of Department)

Dr. Gopi B Sanghani
(Dean)

Attachment



The screenshot displays the CardioML web application interface. At the top, there is a navigation bar with the CardioML logo, links for Data Insights, Model Info, and Disclaimer, and a Predict Now button. The main content area features a large heading "Cardiovascular Risk Assessment" and a subheading "Early Warning System". Below the heading, a description states: "Our AI model analyzes clinical variables to predict potential heart risks with precision." There are two buttons: "Start Assessment" and "Learn More". To the right, there are four feature cards: "73.4% Accuracy" (Trained on 70,000+ clinical patient records with cross-validation), "Instant Results" (Real-time gradient boosting classification in under 5 seconds), "Secure & Private" (End-to-end encrypted processing with anonymous data), and "Confidence Scores" (Predictions include probability-based confidence levels).

CardioML Data Insights Model Info Disclaimer Predict Now

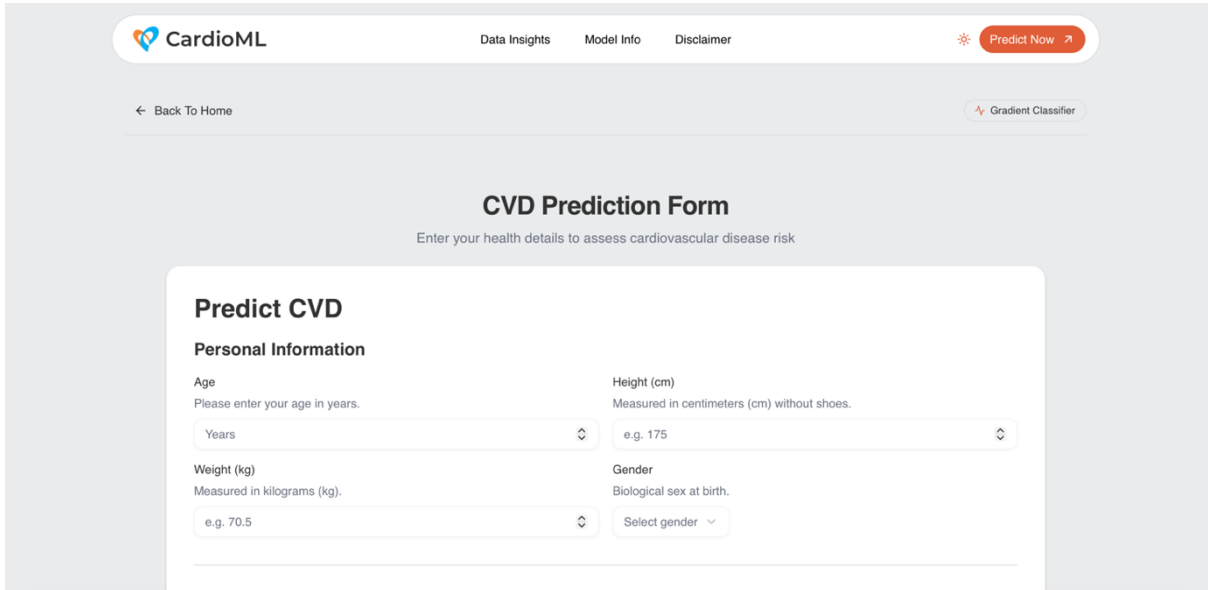
Early Warning System

Cardiovascular Risk Assessment

Our AI model analyzes clinical variables to predict potential heart risks with precision.

Start Assessment Learn More

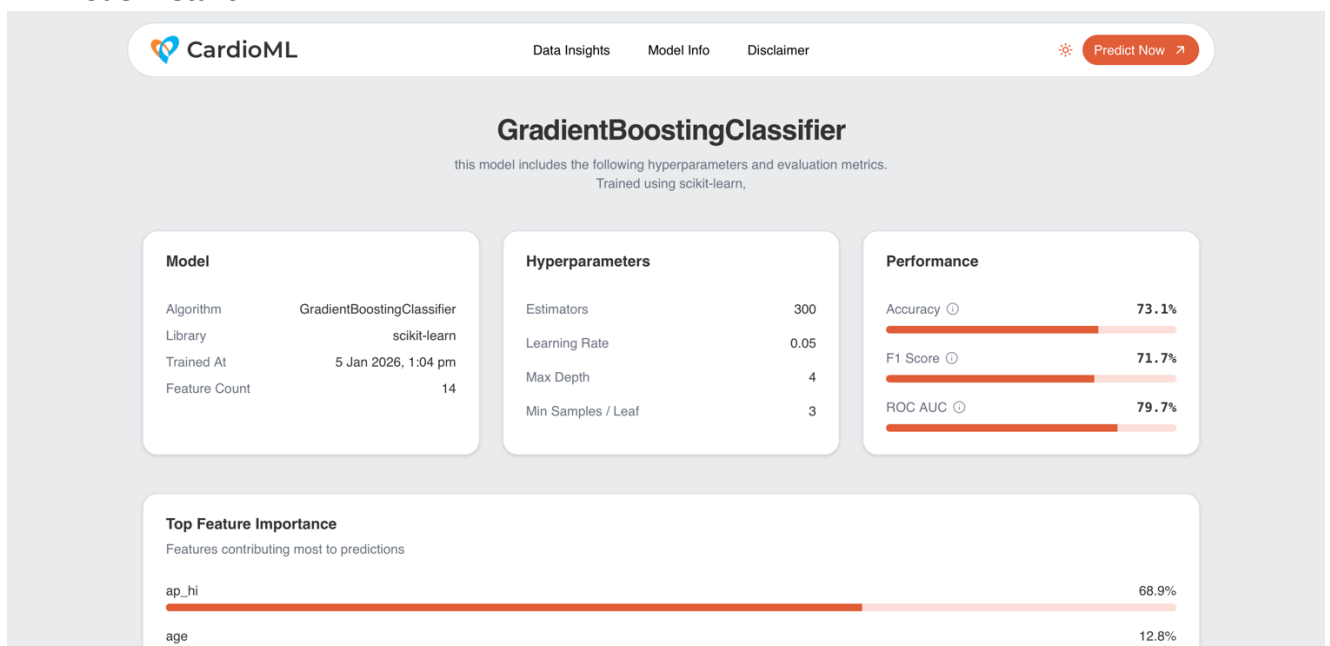
- 73.4% Accuracy**
Trained on 70,000+ clinical patient records with cross-validation.
- Instant Results**
Real-time gradient boosting classification in under 5 seconds.
- Secure & Private**
End-to-end encrypted processing with anonymous data.
- Confidence Scores**
Predictions include probability-based confidence levels.



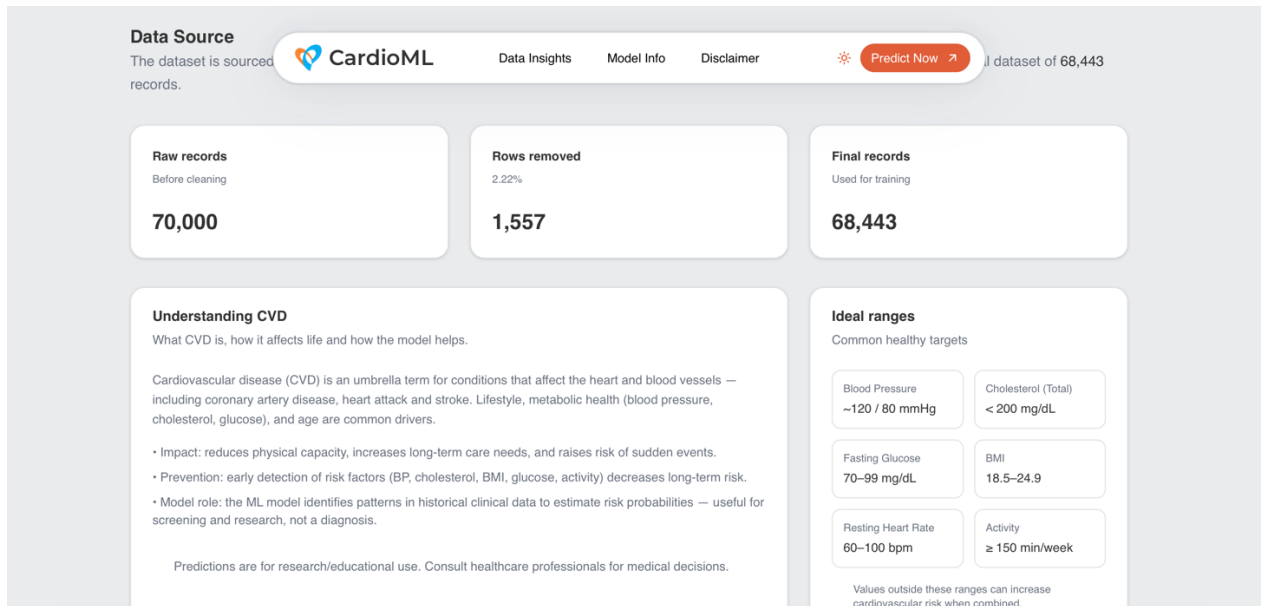
The screenshot shows the 'CVD Prediction Form' on the CardioML website. The form is titled 'Predict CVD' and is for 'Personal Information'. It includes input fields for Age (in years), Height (in cm), Weight (in kg), and Gender (biological sex at birth). There are also links for 'Data Insights', 'Model Info', 'Disclaimer', and a 'Predict Now' button. A 'Back To Home' link is also present.

Prediction Screen

Model Details



EDA Details



Best of luck!

